Operational Technology Graduate Demonstration Project Definition

Project Definition Brief

Project Overview

This project simulates a small graduate engineering work package within a fictional Operational Technology (OT) consulting engagement for an electricity distribution utility.

The fictional client is undertaking an upgrade to its Distribution Control Environment (representing systems such as SCADA, ADMS and related operational platforms). As part of this upgrade, a graduate engineer has been assigned responsibility for validating a small section of the upgraded operational workflow before it is accepted for commissioning.

The project demonstrates the type of engineering thinking, investigation, testing, validation and technical communication that may be expected from a graduate contributing to an Operational Technology consulting project.

The Fictional Client

The project will create a fictional electricity distribution company (working name **TasGrid** for now).

The company operates a simplified distribution network consisting of:

- one zone substation
- two primary feeders
- several distribution sections
- a normally-open tie switch
- several customer zones
- simplified electrical loads
- simplified telemetry points

The network is intentionally small enough to understand completely while still representing the concepts found in a real distribution system.

No real TasNetworks or other utility data will be reproduced.

The Fictional Project

The fictional client is upgrading part of its Distribution Control System.

The upgraded environment includes:

- improved operational workflows
- updated network model
- new automation logic
- improved event logging
- better operational visibility

Before commissioning, a number of validation activities must be completed.

The graduate engineer has been assigned one validation package.

This project represents that package.

Graduate Responsibilities

Within the fictional project, the graduate engineer is responsible for:

Understanding

- the simplified electricity network
- operational workflows
- information flow between operational systems
- engineering assumptions

Validation

- executing engineering test cases
- comparing expected and observed behaviour
- recording results
- identifying failures

Investigation

- analysing unexpected behaviour

- tracing configuration and data issues
- determining root causes
- recommending corrections

Verification

- correcting the identified issue
- repeating validation
- confirming successful operation

Documentation

- recording engineering assumptions
- documenting validation evidence
- writing defect reports
- maintaining traceability
- communicating findings

Operational Systems Represented

The project will simulate interactions between several operational technology systems.

These are conceptual representations only.

System	Purpose within the project
SCADA	Provides telemetry, breaker status, alarms and measurements
ADMS	Maintains network state and operational decision support
OMS	Represents outage information and customer impacts
GIS	Defines network connectivity and asset relationships
Corporate Systems	Stores event history and engineering records

The objective is to demonstrate understanding of how these systems interact rather than replicate commercial software.

Engineering Scenario

The project centres around a feeder fault occurring on one section of the network.

The simulation demonstrates how the upgraded operational environment should respond.

The expected workflow is:

Fault occurs

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Protection trips feeder breaker

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Network topology updates

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Downstream section becomes de-energised

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Alarm generated

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Outage identified

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Customers affected calculated

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Operator acknowledges alarm

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System proposes restoration

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Capacity check performed

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Restoration permitted or rejected

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Actions logged

Validation Scenario

The project validates that the upgraded system behaves correctly.

Validation includes checking that:

- alarms are generated correctly
- topology updates correctly
- affected customers are calculated
- restoration only occurs where electrically valid
- simplified engineering constraints are respected
- operational events are recorded correctly

Engineering Defect Investigation

One intentional engineering defect will be included.

Examples currently under consideration include:

- incorrect switch normal state
- incorrect topology connection
- stale telemetry
- inconsistent configuration data

The graduate engineer will:

- identify the failed validation
- investigate the cause
- locate the incorrect configuration
- apply the correction
- repeat testing
- verify successful operation

Simplified Engineering Analysis

The project includes one simplified engineering calculation supporting restoration decisions.

Rather than performing a full power-system study, the project will demonstrate basic engineering reasoning by checking:

- feeder loading
- transferable load
- alternate feeder capacity
- resulting loading percentage

This provides an engineering basis for restoration decisions while remaining within the project's educational scope.

Demonstrator

The demonstrator is a visual representation of the engineering work package.

It will allow reviewers to:

- understand the network
- execute validation scenarios
- inspect operational behaviour
- investigate failures
- observe corrections
- review engineering documentation

The interface exists to support the engineering process rather than to showcase frontend development.

Supporting Documentation

The software is only one component of the project.

The complete project will also include:

- Engineering Design Brief
- Operational Technology research notes
- Requirements Specification
- Network Model

- Operational Workflow
- System Context Diagram
- Data Flow Diagram
- Requirements Traceability Matrix
- Validation Test Plan
- Validation Results
- Defect Investigation Report
- Engineering Assumptions
- Final Evaluation

Each document supports the same engineering scenario and forms part of a coherent client-delivery package.

Learning Objectives

The project is designed to develop practical understanding of:

Electrical Engineering

- distribution fundamentals
- feeders
- substations
- switching
- simplified loading

Operational Technology

- SCADA
- ADMS
- OMS
- GIS
- operational workflows
- system integration

Engineering Delivery

- requirements engineering
- testing and validation
- defect investigation
- engineering documentation
- traceability
- configuration management

Software Engineering

- modelling operational systems
- database-backed workflows
- visualisation
- engineering-focused UI
- validation tooling

Final Deliverable

The completed project should resemble a **small, self-contained engineering work package** that could plausibly have been completed by a graduate engineer working under the supervision of senior consultants during an Operational Technology upgrade project.

It is not intended to demonstrate mastery of proprietary utility software or advanced power-system analysis.

Instead, it demonstrates:

- structured learning,
- practical engineering reasoning,
- systems thinking,
- operational technology awareness,
- validation methodology,
- technical communication,
- and software implementation within the context of an electricity distribution consulting engagement.
